

Abstract

Public land-cover products are increasingly used where authoritative local time series are unavailable, but product choice and within-product discontinuities can alter apparent urban change and model rankings. We present an auditable protocol for constrained spatial allocation using Geospatial Kernel, the algorithmic core of a Geospatial World Model, as the focal execution layer. Two annual 10-m products for Abu Dhabi city, Google Dynamic World (2017–2024) and an ArcGIS-served Impact Observatory/Microsoft/Esri series (2017–2025), were harmonized to the same 100-m, six-class modelling contract. Leakage-controlled expanding-window tests compared a FLUS-style ANN-CA, Geospatial Kernel, GeoFM-LDN, persistence and random minimum-change allocation using three seeds and 1,000 spatial-block bootstrap resamples. The target year supplied oracle class totals and evaluation labels only. In 2024, the ArcGIS-served series mapped 335.57 km² as built versus 155.98 km² in Dynamic World; built-class intersection over union was 0.449. Its built class also rose by 30.6% between 2021 and 2022 while wetland and woody classes later nearly disappeared, an unreconciled series break. Kernel had the highest mean FoM in four Dynamic World targets and two matched ArcGIS-served targets; these numerical ranks did not uniformly imply paired-interval separation. Net-change allocation imposed product-specific count-only FoM upper bounds of 0.086–0.813 only for exact minimum-change allocators. Kernel's ArcGIS-minus-Dynamic World strict change Figure of Merit ranged from -0.244 to +0.205. The product-specific 2031 frontiers retained FLUS-style and Kernel candidates in all scenarios, but their objective trade-offs changed. The results support an inspectable proposal–projection–writeback boundary while showing that comparative skill is label-product dependent. Neither product is authoritative local land-use truth, model outputs remain 100 m, and the 2026–2031 maps are conditional stress tests rather than official forecasts.